

PROJECT DESIGN PHASE-II

Solution Requirements Specification (SRS)

Date	18 June 2026	Project Name	ShopEz (MERN Full Stack)
Team ID	--	Maximum Marks	4 Marks
Student	Turlapati Lahari (24eg109a54@anurag.edu.in)		

Functional Requirements (FR)

ID	Requirement (Epic)	Description & Logic
FR-1	User Signup - Data Entry - Verification	Enables platform access for new users via email registration and encrypted credential storage.
FR-2	User Access Control - JWT Handling - Logout	Implements token-based stateless security for session management and authenticated routing.
FR-3	Categorized Items - Real-time Feed - Catalog View	Displays dynamic lists for Men, Women, and Kids segments sourced from MongoDB.
FR-4	Search Functionality Keyword Matching	Allows shoppers to browse the warehouse using text queries with immediate results.
FR-5	Client-Side Filters - Cost - Type - Size	Features instant sorting and filtering without full-page reloads using React logic.
FR-6	Cart Coordination - DB Sync - CRUD Ops	Facilitates adding, modifying, or removing items in a persistent shopping bag.
FR-7	Checkout Pipeline - Shipping - Mock Pay	Captures user logistics and provides a summary for secure finalization.
FR-8	Order Monitoring Status Tracking	Portal for buyers to check real-time progress (Placed/Transit/Complete).
FR-9	Admin Catalog Tools - Assets - Price Edits	Private area for managing store stock, including media uploads and price tweaks.
FR-10	Admin Operations - Logistics - Trends	Centralized dashboard to observe global transactions and evaluate performance.

Non-Functional Requirements (NFR)

ID	Attribute	Requirement Specification	Importance
NFR-1	Security	All credentials hashed with Bcrypt; Private routes shielded via JWT logic.	CRITICAL
NFR-2	Performance	Initial interaction < 2s; API result delivery for queries < 500ms.	HIGH
NFR-3	Scalability	System must support horizontal data distribution using NoSQL partitioning.	HIGH

ID	Attribute	Requirement Specification	Importance
NFR-4	Responsiveness	Context-aware layout ensuring high usability across mobile and desktop.	HIGH
NFR-5	Availability	Targeting 99.9% operational uptime during high-traffic intervals.	CRITICAL
NFR-6	Data Integrity	Consistent relationships enforced between Accounts, Products, and Orders.	CRITICAL